

Medical Insurance Claim Fraud Detection

Approach 2: Deep Learning with Explainable AI

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Abstract

This report presents a deep-learning extension to a reproducible medical-insurance claim fraud baseline. The same repository-provided workbook, feature transformations, and 70/15/15 split used by the traditional approach are retained for fair comparison. Five architectures—an MLP, TabNet-style attentive network, one-dimensional CNN, autoencoder anomaly hybrid, and feature-token transformer—are trained with BCE-with-logits, inverse-frequency fraud weighting, AdamW, gradient clipping, checkpoints, and early stopping. Three fixed seeds quantify stability. The shared evaluator reports F2, PR-AUC, ROC-AUC, calibration, fairness, and test intervals. Explainability is treated as a first-class output: all models receive comparable occlusion importance, deletion-faithfulness, and jitter-stability analyses, while native masks and attention views remain auxiliary. The results are limited by the 4,500-row supplied workbook, missing policy/document fields, and possible synthetic shortcuts. The deep approach is therefore a controlled research comparison, not a production claim-denial system.

Keywords: deep learning; fraud detection; health insurance; tabular learning; explainable AI; India

1. Introduction

Traditional models provide a strong baseline for structured claims. Deep learning is evaluated because learned interactions and representation capacity may help when categorical and temporal evidence becomes richer. The objective is not to assume neural superiority but to measure whether its additional capacity is justified.

2. Dataset comparability

The deep run consumes 4,500 claims with 270 fraud labels and the same frozen feature registry. The input checksum is 952b6e23c9845b9086994c08bc81323575968c318f845ea69e3b9609614b6a45; the split fingerprint is 3bc2230ff033d29d4eb664ca8b78f2c902416a154bcc0cee88d0863881d59c9f. IDs and likely post-decision fields remain excluded.

3. Architecture designs

The MLP provides a controlled dense baseline. The TabNet-style model applies step-specific soft masks. The CNN tests local feature neighborhoods while documenting order sensitivity. The autoencoder reconstructs rows and supplies an anomaly feature to a supervised head. The transformer treats each scalar column as a token and uses self-attention.

4. Training protocol

All models use the same seed list, batch contract, optimizer family, weighted loss, gradient clipping, telemetry, checkpoint restoration, and early stopping rule. The reference run uses a laptop-sized three-configuration budget; larger Bayesian searches are explicitly future work rather than fabricated results.

5. Evaluation and XAI

Mean validation probabilities across seeds select the threshold. The test set is evaluated only after the leaderboard and XAI assets are complete. Occlusion importance, faithfulness, and stability are common scored methods; native masks, tokens, and reconstruction contributions are explanatory context.

6. Results

The selected deep model is Feature-token transformer with validation PR-AUC 0.9800, validation F2 0.9709, and test F2 0.9756. The full leaderboard remains in `evaluation2/leaderboard.csv`.

7. Fairness and ethics

Demographic columns are used for audit slices, not direct denial features. Small groups are marked unstable. A deep model is not allowed to infer claimant intent; a flagged result routes to document and policy review.

8. Limitations and future work

The input is still a transformed tabular matrix. Future research should add entity embeddings, temporal sequences, graph structure, document evidence, external validation, and calibrated human-in-the-loop review.

9. Deep leaderboard

Rank	Model	Val PR-AUC	Val F2	PR std	Test F2
1	Feature-token transformer	0.9800	0.9709	0.0058	0.9756
2	TabNet-style attentive network	0.9782	0.9709	0.0038	0.9662
3	1D convolutional tabular network	0.9763	0.9570	0.0033	0.9479
4	MLP	0.9748	0.9662	0.0047	0.9662
5	Autoencoder anomaly hybrid	0.9528	0.9362	0.0210	0.2395

Appendix A — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 01 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix B — Deep-learning audit exhibit

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Exhibit 02 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix C — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 03 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix D — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 04 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix E — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 05 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix F — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 06 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix G — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 07 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix H — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 08 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix I — Deep-learning audit exhibit

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Exhibit 09 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix J — Deep-learning audit exhibit

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Exhibit 10 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix K — Deep-learning audit exhibit

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Exhibit 11 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix L — Deep-learning audit exhibit

This appendix records the evidence chain for architecture, seed, telemetry, explanation, fairness, calibration, and selection. It is generated from the structured run context. The numbers are not hand-edited and should be traced to the corresponding JSON or CSV artifact before a defense.

Exhibit 12 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix M — Deep-learning audit exhibit

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Exhibit 13 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix N — Deep-learning audit exhibit

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Exhibit 14 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix O — Deep-learning audit exhibit

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Exhibit 15 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix P — Deep-learning audit exhibit

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Exhibit 16 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix Q — Deep-learning audit exhibit

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Exhibit 17 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix R — Deep-learning audit exhibit

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Exhibit 18 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix S — Deep-learning audit exhibit

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Exhibit 19 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix T — Deep-learning audit exhibit

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Exhibit 20 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix U — Deep-learning audit exhibit

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Exhibit 21 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix V — Deep-learning audit exhibit

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Exhibit 22 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix W — Deep-learning audit exhibit

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Exhibit 23 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix X — Deep-learning audit exhibit

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Exhibit 24 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix Y — Deep-learning audit exhibit

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Exhibit 25 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

Appendix Z — Deep-learning audit exhibit

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Exhibit 26 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

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Exhibit 27 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

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Exhibit 28 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

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Exhibit 29 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

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Exhibit 30 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.

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Exhibit 31 evidence paths: ``evaluation2/metrics/``, ``evaluation2/xai/``, ``evaluation2/fairness/``, ``images2/``, ``documentation2/complete_deep_learning_manual.md``. The same target semantics and reviewer-first boundary apply to this exhibit.